

kea: A Python package of tools for consistent power spectral analysis of data on arbitrary dimensions

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DOI: [10.xxxxxx/draft](https://doi.org/10.xxxxxx/draft)

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Submitted: 01 January 1970

Published: unpublished

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Summary

Power spectral density techniques are widely used across disciplines but often differ in normalization and data handling, with most fields relying on a limited set of established methods. Historically, researchers have been required to implement their own routines or personally request code from others. This increases barriers to entry for performing analysis and requires a deep understanding of specific Fourier conventions to avoid errors. Without a standardized framework, order-unity normalization differences can creep in, which can lead to physical inferences that are mathematically inconsistent or physically erroneous. To address this, we present a computational package that provides consistently normalized estimators that are adaptable to multiple data types – including intracluster medium surface brightness fluctuations, in-situ solar wind time-series, and simulation cubes.

Statement of Need

Autocorrelation functions (ACF), power spectral densities (PSD), and structure functions (SF) serve as the primary mathematical framework for quantifying the statistical properties of stochastic phenomena across varying spatial and temporal scales. In the context of fluid dynamics and astrophysics, these tools allow researchers to decompose complex, multiscale signals into their constituent parts. Of particular interest, especially within the study of (magneto)-hydrodynamic turbulence, are the characteristic scales (such as the integral scale L where energy is injected, and the dissipation scale η), the power-law indices that define the energy cascade (e.g., the Kolmogorov $-5/3$), and their respective amplitudes which dictate the total turbulent energy budget.

These phenomena are rarely captured in the full continuum of values in the available space in which measurements occur. In other words, in a lot of practical cases, observations are restricted to only 1D slices, 2D projections, or 2D slices. For example, in the case of the solar wind, in-situ measurements are performed by sensors moving in relation to the plasma rest frame. These sensors generate a time-series of data: such as magnetic field (vector field), or density (scalar field) measurements. As another pertinent example, surface brightness observations of the intracluster medium (ICM) observe 2D (emission weighted) projections of scalar fields, or projections of line-of-sight components of a vector field. Often these observations are calibrated or compared against more accessible regimes, or high-fidelity simulations.

The development of robust astrophysical and fluid models to explain these phenomena is inherently an iterative process. As observational data reach higher sensitivities and angular resolutions, researchers increasingly rely on high-fidelity, simulations to probe unobservable

41 dynamics and generate artificial observations. Meaningfully comparing these complex
42 simulations against lower-dimensional, real-world observations necessitates a consistent,
43 universal, and dimensionally-adaptable approach.

44 Beyond the choice of statistical tool, the reliability of the output is heavily dependent on the
45 nuances of spectral estimation. There exists a vast library of signal processing techniques
46 designed to mitigate the distortions, aliasing, and biases inherent in finite sampling (Maciejewski
47 et al., 2009; Sefusatti et al., 2016; Stoica & Moses, 2005). However, the implementation of
48 these techniques introduces its own set of variables: how the data are windowed to prevent
49 spectral leakage, how the estimates are binned in k -space, and how the resulting power is
50 normalized (e.g., ensuring Parseval's theorem is satisfied). For example, there are a total of six
51 different R packages that provide spectrum estimations with varying normalization options
52 (Barbour & Parker, 2022).

53 If these methodological choices are not standardized, they can induce spurious physical behavior.
54 For example, improper binning can artificially flatten a spectral slope. This could become a
55 significant issue when comparing across different studies or when cross-correlating observational
56 data with numerical models. Without a rigorous, consistent approach to these estimations,
57 the resulting physical interpretations (such as the injection or dissipation scales) may reflect
58 the limitations of the signal processing rather than the underlying physics.

59 We thereby introduce Kea, a Python package that implements several different dimensionally-
60 agnostic scale-dependent statistical estimation techniques for use in turbulence analysis. Kea is
61 available on GitHub via <https://github.com/mab68/Kea>.

62 Power-law correlations are a key feature of turbulence (particularly with high Reynolds numbers
63 flows). The scale-to-scale energy transfer associated with the cascade of energy from large
64 scales to smaller scales leads to a power-law power spectrum which is indicative of fractal-like
65 structure where small-scale features are statistically similar to large-scale ones.

66 Observing the complete set of information provided by self-similar physical phenomena can
67 be practically challenging. In particular, we highlight that the majority of astronomy and
68 astrophysical observations are driven by the detection and manipulation of photons. The
69 properties of these photons are determined by the conditions of the source (temperature, density,
70 velocity, etc.) of the emitting material and its subsequent interaction with any intervening
71 material. Determining the conditions of the source from the photons alone can be a serious
72 challenge. Noise and resolution constraints may also limit the accuracy of the inference
73 and increase complexity. To understand biases and systematics or comparison of theory to
74 observation, where data is lacking, it can be necessary to compute expected observational
75 properties from controlled surrogate models (Haworth et al., 2018; Simionescu et al., 2019).
76 Kea additionally provides dedicated utilities for generating synthetic spatial fields.

77 State of the Field

78 While the mathematical definitions of autocorrelation functions, power spectral densities
79 and structure functions are well-established, their application to “real-world” datasets show
80 significant variability in subtle ways. For example, normalizations of the difference-of-Gaussian
81 PSD method differ (Arévalo et al., 2012; Churazov et al., 2012; Zhou et al., 2022). Additionally,
82 the exact normalization factors change depending on the chosen PSD representation (e.g.,
83 angle-averaged, angle-integrated, or amplitude spectra). To resolve these inconsistencies, Kea
84 provides dedicated routines for conversion between both these spectral representations and
85 their corresponding normalization conventions.

86 Currently available software for ICM turbulence analysis are: turbustat (Koch et al., 2019), and
87 PITSZI (Adam et al., 2025). Both of these solutions require 2D data, thus are not available
88 for applications to e.g., solar wind turbulence analysis or simulation cubes. Additionally,
89 they both use a nested sequence of object-oriented class based design with built-in handling

with assumptions for their respective fields: the interstellar medium for turbustat, and the intracluster medium for PITSZI. Hence, modifying for individual or specific needs is challenging.

For SF calculations, fastSF (Sadhukhan et al., 2021) supports 2D and 3D datasets but strictly requires uniform grids, rendering it incompatible with gapped or irregularly sampled data. While other Python packages like fluidsSF (Wagner et al., 2025) and PyTurbo_SF (Ayouche et al., 2026) offer SF calculations, they lack a unified framework that couples arbitrary dimensionality handling with gapped-data support.

Finally, tools like powerbox (Murray, 2018), GaussianRandomFields.jl (Robbe, 2023), and DRDMannTurb (Izmailov et al., 2024) generate synthetic turbulence or Gaussian random fields, they are largely decoupled from the analytical toolkits required to seamlessly validate empirical estimators against these synthetic baselines.

Software Design

Kea is designed with two main goals: provide standardized tools for robust, multidimensional spectral estimation, and to generate controlled synthetic fields for validating these analytical methods. To efficiently support these objectives, Kea is written as a Python package that operates directly on NumPy arrays, ensuring it is easily integrable across varying fields of research and workflows. Kea additionally has optional support for GPU computation using CuPy for structure function, autocorrelation function, and difference-of-Gaussian PSD estimation.

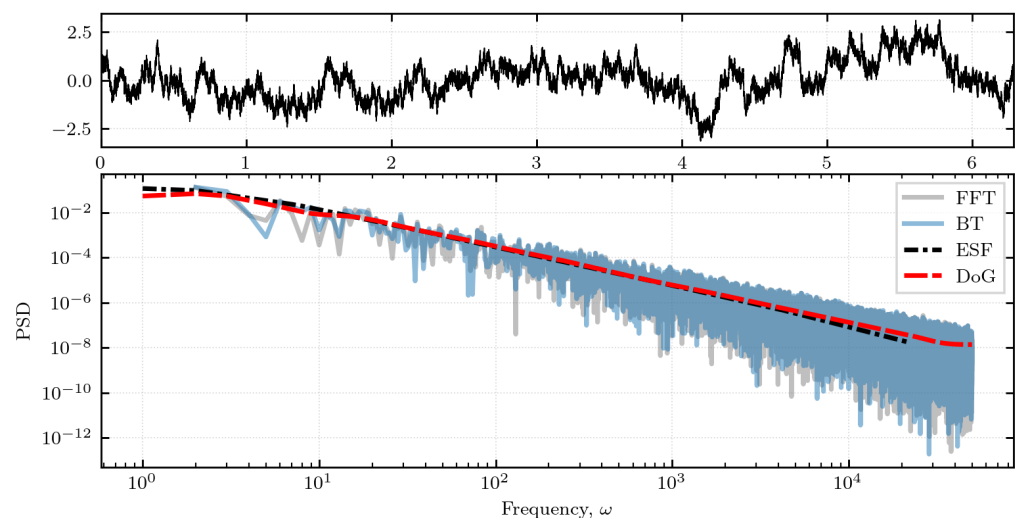


Figure 1: Spectral estimation capabilities of Kea. The top panel displays a 1D synthetic timeseries generated by the software, while the bottom panel illustrates the resulting spectral estimates computed using Kea's implementations of the periodogram (FFT), Blackman-Tukey (BT), equivalent spectrum (ESF), and difference-of-Gaussian (DoG) methods.

Kea provides discrete implementations of several key spectral estimation techniques. An application of these methods to a synthetic timeseries is demonstrated in Figure 1. Since the autocorrelation function (ACF), structure function (SF), and the FFT and BT spectral estimates yield D -dimensional functions, it is often necessary to bin them to reduce their dimensionality. To address this, Kea provides a general binning routine applicable to both the spectral estimates (PSD) and the lag-functions (ACF, SF).

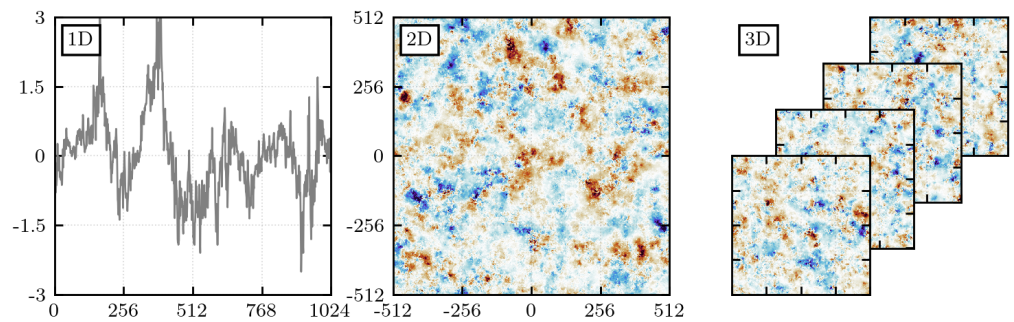


Figure 2: Synthetic multifractal test fields generated by Kea, demonstrating the package's support for 1D (left), 2D (middle), and 3D (right, shown as a 2D slices) spatial data structures. These generation tools allow users to easily create controlled datasets for testing and benchmarking.

114 Synthetic fields have been used extensively to validate methods and measure turbulence in
115 the interstellar medium (Brunt & Heyer, 2002; Esquivel et al., 2003; Miville-Deschênes et
116 al., 2003; Ossenkopf et al., 2006) and similarly for the ICM (Vogt & EnBlin, 2005; XRISM
117 Collaboration et al., 2025; ZuHone et al., 2016). Kea provides methods to produce (monofractal
118 and multifractal) synthetic fields with correlations parameterized by power spectral density.
119 These fields are synthesized in Fourier-space and Fourier-transformed to provide real-space
120 fluctuation fields (Barnsley et al., 1988; Lakhali et al., 2025).

121 For example, shown in Figure 2 are one, two, and three-dimensional multifractal fields. Thus,
122 the below PSD estimators can be tested for validity with known/expected forms, and their
123 performance tested under constraints of noise and missing data.

124 Research Impact Statement

125 Kea bridges the gap between empirical spectral estimation and synthetic field generation,
126 providing a unified framework for researchers analyzing turbulence and continuous spatial data.
127 By operating directly on standard NumPy arrays without rigid, domain-specific assumptions, Kea
128 serves as an accessible, field-agnostic toolkit for calculating scale-dependent statistics (ACFs,
129 PSDs, and SFs) across 1D, 2D, and 3D datasets.

130 Crucially, Kea enables researchers to rigorously model complex observational biases – such as
131 noise and missing data gaps – by generating controlled, stochastic monofractal and multifractal
132 fields. This integrated approach allows users in fields ranging from astrophysics to fluid
133 dynamics to robustly benchmark and validate spectral estimators against synthetic ground
134 truths before applying them to messy, real-world observational data.

135 In summary, Kea offers the following:

- 136 • **Domain-agnostic & Lightweight Design:** Built directly on NumPy arrays making it easily
137 integrable into existing workflows across astrophysics, fluid dynamics, and spatial analytics.
138 Has optional additional GPU support with CuPy.
- 139 • **Flexible Dimensionality Support:** Native handling for 1D time series, 2D image data,
140 and 3D simulation cubes.
- 141 • **Gapped Data Compatibility:** Provides methods for estimation of structure functions,
142 autocorrelation functions, and power spectral densities on datasets with gaps represented
143 by NaN.
- 144 • **Unified Analytical Toolkit:** Provides discrete implementations for periodogram, Blackman-
145 Tukey, equivalent spectrum, and difference-of-Gaussian spectral estimates.
- 146 • **Normalization & Representation Standardization:** Offers built-in conversion routines
147 between different PSD binning methods and normalizations.

- **Synthetic Field Generation:** Synthesizes controlled D -dimensional stochastic fields (both monofractal and multifractal) parameterized by PSDs to create ground-truth baselines for benchmarking algorithms.

Comprehensive documentation is available at <https://astrokea-docs.readthedocs.io/en/latest/index.html> with tutorials, and practical examples. The documentation additionally details the discretized scale-dependent statistical estimations, binning routines, and synthetic fields.

AI Usage Disclosure

The paper was initially drafted in full by the authors. Google Gemini 3.1 was used after the drafting of this manuscript to assist with linguistic polishing to improve manuscript clarification. Google Gemini 3.1 was also used to suggest improvements to the code. All AI-assisted text and code were thoroughly reviewed, tested, and verified by the authors to ensure correctness. The authors assume full responsibility for the final content and scientific accuracy of the manuscript and code.

Future Work

It is intended for future development of kea to implement a general interpretation of spectral estimation using arbitrary filters along with automatic implementation of debiasing using the non-parametric local power-law approximation that is currently only available for the ESF method.

Another area of potential improvement will be the implementation of synthetic fields with anisotropy. This would be useful to test the statistics of projected fluctuations in more complex scenarios.

Acknowledgements

This project was supported by the Marsden Fund Council from New Zealand Government funding, managed by Royal Society Te Apārangi (No. E4200).

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